Chain-Aware Augmentation Mixing for Corruption-Robust Continual Learning

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Abstract

How does the share of the original image in an augmented view affect corruption robustness in continual learning? We compare chain-aware mixing and a recurrent MetaMix baseline at five original-image weights on Split-CIFAR-10. Across three seeds, classifier InfoNCE improves mean corrupted class-incremental accuracy over Jensen–Shannon (JS) consistency at all five weights for chain-aware mixing and at four for MetaMix. At the strongest tested augmentation ($m = 0.1$), the two InfoNCE classifiers reach 43.89% and 41.66%, respectively. At $m = 0.3$ their order reverses (42.28% and 42.51%). The AugMix reference leads on corrupted task-incremental accuracy at 84.15%. The results link classifier consistency to mixture strength and show why conclusions about mixer design depend on which metric and strength are compared.

1. Introduction

Serena: Continual learning methods typically mitigate catastrophic forgetting through parameter regularization [**?**], architectural expansion, or experience replay [1**?**], yet replay-based models remain brittle under distribution shifts and common corruptions [2]. Although automated data augmentation [3**?**] and supervised contrastive learning [4] improve out-of-distribution robustness, meta-learning augmentation policies [5] can lead to trivial solutions when the mixer satisfies consistency simply by weighting the clean image.

Prediction consistency can be improved by weakening augmentation rather than learning robustness: if mixed views approach the original image, their disagreement vanishes. In continual learning, this choice also affects updates shared by current and replayed examples. AugMix samples augmentation mixtures [3]; MetaMix adapts their weights from recurrent learner context [5], but its weights do not depend on the sampled chains' effects.

We study chain-aware mixing at fixed augmentation strengths. Our mixer scores each candidate using its feature and prediction changes, and learns through a classifier update evaluated on independently mixed pseudo-unseen views. Fixing the original-image weight prevents the mixer from reducing consistency loss simply by suppressing augmentation. Independent outer mixtures ensure that its feedback measures the effect on classifier learning.

Our contributions are:

- A chain-aware mixer that combines recurrent learner context with candidate descriptors to weight chains separately for each image and seen view.
- A differentiable one-step training objective whose outer loss reaches the mixer exclusively through the classifier update.
- A three-seed comparison across five mixture strengths, showing that the benefits of mixer design and classifier consistency depend on strength and evaluation metric.

2. Methodology

Views and mixer inputs. Combine current and labeled replay batches into (x_b, y_b). Disjoint pseudo-seen and pseudo-unseen operation pools produce two views each, denoted x'_{b1}, x'_{b2} and $\hat{x}_{b1}, \hat{x}_{b2}$. Each view mixes three sampled chains:

$$\tilde{x} = mx + (1-m)\sum_{k=1}^{3} w_k a_k(x), \quad w_k \geq 0, \quad \sum_k w_k = 1. \tag{1}$$

Larger m means weaker augmentation. Before updating classifier parameters θ, both mixers encode detached context

$$I_i = [g_t^\top g_r,\ L_r,\ e_{r,i},\ e_{t,i}], \tag{2}$$

where g_t, g_r are clean current/replay loss gradients, L_r is replay cross-entropy, and $e_{t,i}, e_{r,i}$ are pooled features. Gradient alignment indicates local compatibility between current and replay updates.

MetaMix maps the LSTM state to weights shared across views and pools; it either learns m or uses a fixed value.

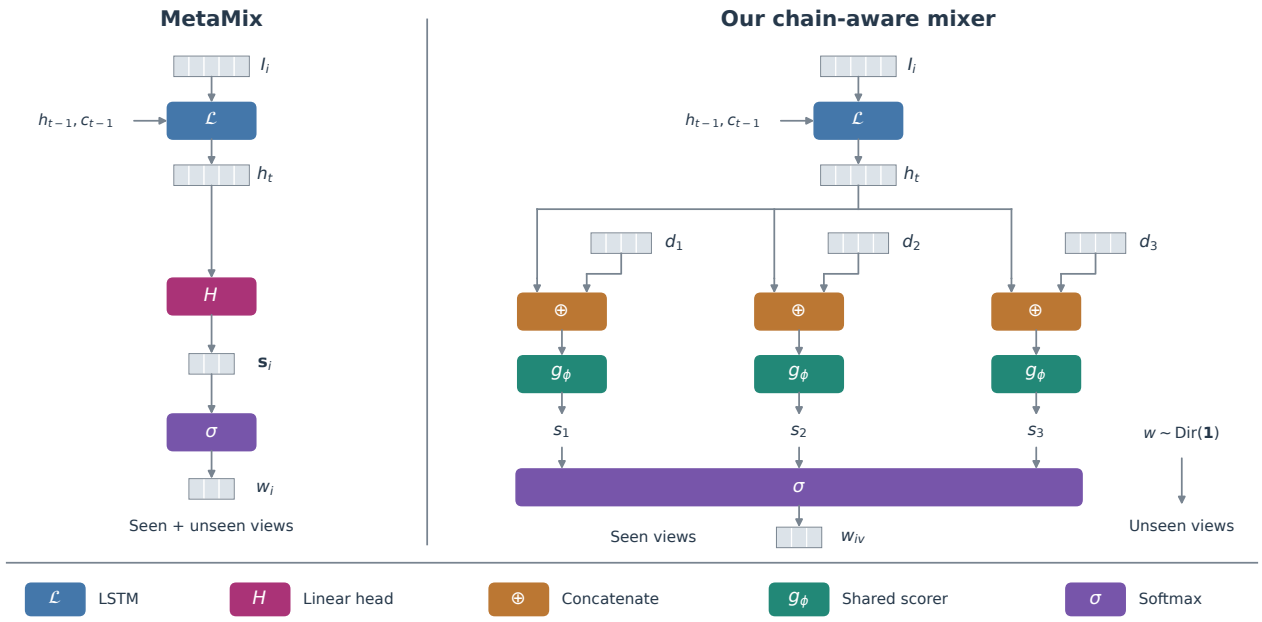


Figure 1. MetaMix and our chain-aware mixer. Colors and symbols identify modules in the legend; cell arrays are schematic. Both mixers encode learner context I_i with an LSTM. MetaMix predicts weights shared across views and operation pools. Our shared scorer independently maps each pair (h_t, d_k) to a scalar s_k; softmax then combines the three scores into weights for each seen view. The detached descriptor $d_k = [\Delta e_k, \Delta p_k]$ records changes in pooled features and class probabilities relative to the clean image. Pseudo-unseen weights are sampled independently. The original-image mixing branch is omitted.

Our mixer uses fixed m and independently scores each detached descriptor $d_k = [\Delta e_k, \Delta p_k]$, containing feature and probability changes relative to the clean image. A shared scorer combines d_k with the current hidden state h_t; softmax converts the three scalars into per-image, per-seen-view weights (Fig. 1). Pseudo-unseen weights are independent $\mathrm{Dirichlet}(1, 1, 1)$ draws, held fixed during each batch's mixer updates.

What the classifier learns. Write $\ell_\theta(x, y) = \mathrm{CE}(f_\theta(x), y)$ for mean batch cross-entropy, where f_θ returns logits. A separate replay draw (x_d, z_d) supplies stored logits for distillation. Our DER++ loss [1] is

$$L_{\mathrm{DER++}}(\theta) = \ell_\theta(x_t, y_t) + 0.5\,\ell_\theta(x_r, y_r) + 0.2\,\mathrm{MSE}(f_\theta(x_d), z_d). \tag{3}$$

MSE averages over examples and logit coordinates. Replay terms are zero when memory is empty. The classifier objective is

$$L_S(\theta, \phi) = L_{\mathrm{DER++}}(\theta) + C_\theta(x_b, x'_{b1}, x'_{b2}). \tag{4}$$

Here C_θ is either Jensen-Shannon divergence or class-aware InfoNCE; the latter also uses the labels y_b. The dependence on ϕ enters through the mixed views.

For three aligned batches v_0, v_1, v_2 of size n, let p_{ij} denote the classifier's probability vector for example i in view v_j, and let $\bar{p}_i = (p_{i0} + p_{i1} + p_{i2})/3$. We use

$$\mathcal{L}_{\mathrm{JS},\theta}(v_0, v_1, v_2) = \frac{1}{3n} \sum_{i=1}^{n} \sum_{j=0}^{2} \mathrm{KL}(p_{ij} \,\|\, \bar{p}_i). \tag{5}$$

InfoNCE instead regularizes normalized pooled features u_a. Each of the $3n$ views is an anchor; its positives P_a are the other two views of the same image. Negatives N_a are all views of different-label images. Other same-class images and the anchor itself are excluded, avoiding explicit same-class repulsion without imposing same-class attraction. With $\tau = 0.1$, we use

$$\mathcal{L}_{\mathrm{NCE},\theta} = -\frac{1}{3n} \sum_{a} \frac{1}{2} \sum_{b \in P_a} \log \frac{\exp(u_a^\top u_b/\tau)}{\sum_{c \in P_a \cup N_a} \exp(u_a^\top u_c/\tau)}. \tag{6}$$

How the mixer guides representation learning. A virtual classifier step evaluates whether the chosen training views improve clean classification and pseudo-unseen consistency:

$$\theta^+(\phi) = \theta - \alpha \nabla_\theta L_S(\theta, \phi), \tag{7}$$

$$L_U(\theta^+, \phi) = \ell_{\theta^+}(x_b, y_b) + \mathcal{L}_{\mathrm{JS},\theta^+}(x_b, \hat{x}_{b1}, \hat{x}_{b2}). \tag{8}$$

Loss	m	Clean class-IL		Corrupt class-IL		Clean task-IL		Corrupt task-IL	
		CA	MM	CA	MM	CA	MM	CA	MM
JS	0.1	43.99	**45.69**	38.20	**39.23**	86.05	**86.56**	80.48	**80.88**
	0.3	**42.61**	40.78	**36.33**	34.31	**87.56**	86.61	**81.29**	79.82
	0.5	**44.59**	43.51	**36.56**	36.38	86.26	**86.89**	79.14	**80.05**
	0.7	**42.94**	41.69	**34.96**	34.26	**86.65**	86.12	**79.63**	79.07
	0.9	42.87	**45.60**	33.41	**36.34**	85.58	**86.55**	77.64	**78.88**
InfoNCE	0.1	**51.87**	49.25	**43.89**	41.66	**88.46**	87.74	**83.24**	82.17
	0.3	**50.27**	50.14	42.28	**42.51**	**87.87**	87.43	**81.94**	81.85
	0.5	**46.26**	45.71	**37.49**	37.27	**87.12**	**87.12**	80.92	**81.01**
	0.7	**46.08**	45.63	**36.72**	36.20	86.96	**87.07**	**79.64**	79.59
	0.9	43.33	**44.01**	34.18	**34.60**	**86.55**	86.53	**79.43**	79.19

Table 1. Mean final accuracy (%) over seeds 17, 23, and 43. Column pairs compare chain-aware (CA) and MetaMix (MM) at the same original-image weight and classifier consistency loss. For each loss and m, bold marks the higher CA or MM mean within each metric pair. Ties at the displayed precision are both bold.

For $J(\phi) = L_U(\theta^+(\phi), \phi)$, holding sampled inputs, incoming state and outer BatchNorm buffers fixed gives

$$\nabla_\phi J = (D_\phi \theta^+)^\top \nabla_{\theta^+} L_U + \partial_\phi L_U. \tag{9}$$

Our independent outer mixtures make $\partial_\phi L_U = 0$: the mixer can reduce the outer loss only through its effect on the classifier update. This favors training views whose updates retain class information and improve consistency on pseudo-unseen transformations; it does not guarantee robustness to test corruptions. MetaMix can additionally change its outer mixtures directly.

3. Experiments

3.1. Setup

We compare the two mixers at matched original-image weights $m \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. At each weight we cross mixer choice with classifier consistency. Table 1 reports all four evaluation metrics so the reader can compare mixers within a row and follow each mixer as augmentation strength changes.

SmallCNN learns five successive CIFAR-10 class pairs, $(0, 1)$ through $(8, 9)$, with ten epochs per task and a 500-example reservoir. Each batch has up to 32 current examples and up to 32 labeled replay examples. The classifier and mixer use constant SGD learning rates of 0.003 and 0.005. Learned mixers receive three SGD updates per mini-batch, while the classifier receives one committed update; the random AugMix references have no mixer updates. Each mixer step recomputes the virtual classifier from the same starting parameters; the classifier commits the update obtained with the incoming mixer.

Final evaluation uses 10,000 clean test images and 15 CIFAR-10-C corruptions at five severities [2]. Each corruption and severity receives equal weight. Class-IL predicts among all ten outputs. Task-IL restricts prediction to the true class pair. The pseudo-unseen operations used in training are sampled training transformations. CIFAR-10-C supplies only the test measurements.

References and aggregation. DER++ with AugMix samples two views from the full nine-operation pool, with independent $m \sim \mathrm{Beta}(1, 1)$ and chain weights drawn from $\mathrm{Dirichlet}(1, 1, 1)$. We evaluate its original JS coefficient of 12 and a coefficient of one matched to the JS classifiers. Both use the same replay and classifier update budgets. Each mean gives equal weight to seeds 17, 23, and 43. We report sample SD across seeds in Table 5 and compute differences from the displayed means.

3.2. Results

Mixture strength across four metrics. For chain-aware InfoNCE all four means decline as m increases (Table 1, Fig. 2). The largest class-IL difference occurs on corruptions: 43.89% at $m = 0.1$ versus 34.18% at $m = 0.9$. MetaMix InfoNCE reaches its highest class-IL means at $m = 0.3$ and its highest task-IL means at $m = 0.1$. With JS, the highest corrupted class-IL mean occurs at $m = 0.1$ for both mixers, although intermediate values give a less regular ordering.

What the chain-aware design adds. With classifier InfoNCE and $m = 0.1$, chain-aware leads on all four metrics. At $m = 0.3$ MetaMix leads on corrupted class-IL by 0.23 points, while chain-aware leads on the other three metrics. At $m = 0.5$ their clean task-IL means coincide to two decimals. The comparison changes both seen-chain scoring and pseudo-unseen mixture construction.

Interaction with classifier consistency. Table 3 holds the mixer and mixture strength fixed when comparing classifier

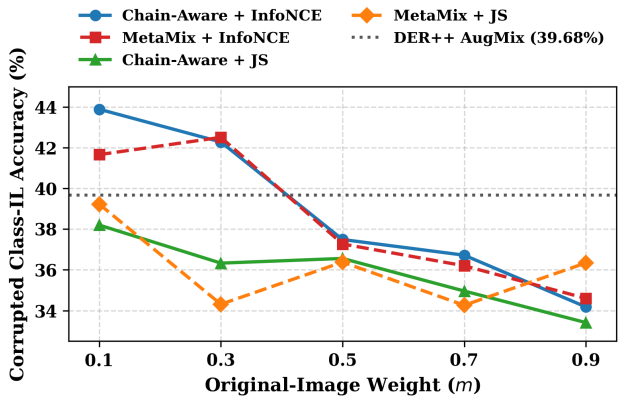


Figure 2. Serena: **Corrupted Class-IL accuracy across original-image weights m.** Three-seed mean accuracy (%) on CIFAR-10-C for Chain-Aware (CA) and MetaMix (MM) under InfoNCE and JS classifier consistency, compared against the matched DER++ AugMix baseline.

losses. Chain-aware gains 5.69 and 5.95 points on corrupted class-IL from InfoNCE at $m = 0.1$ and 0.3, compared with 0.93 at $m = 0.5$. MetaMix gains 8.20 points at $m = 0.3$ but loses 1.74 at $m = 0.9$. The gains also depend on task information: at $m = 0.7$ the chain-aware task-IL gains are only 0.31 points clean and 0.01 corrupted.

Learned and random mixture strength. Tables 1 and 2 show how the fixed settings compare with learned m and random AugMix. Within MetaMix with classifier InfoNCE, corrupted class-IL is 35.26% with learned m, 42.51% at fixed $m = 0.3$, and 34.60% at fixed $m = 0.9$. Serena: It is noted that under unrestricted optimization, the learned m baseline rapidly collapses toward the clean image ($m > 0.9$), yielding a 35.26% corrupted Class-IL accuracy that closely mirrors the fixed $m = 0.9$ boundary (34.60%) and empirically confirms the trivial consistency shortcut. Thus the value chosen matters even when m is fixed. AugMix with its original JS coefficient reaches the highest corrupted task-IL mean (84.15%). Reducing that coefficient to one raises clean class-IL from 45.94% to 48.99%, while corrupted class-IL changes from 40.00% to 39.68%. At the matched coefficient AugMix exceeds both mixers' best observed JS means on corrupted class-IL.

		Class-IL		Task-IL	
Mixer	Loss	Clean	Corrupt	Clean	Corrupt
MetaMix, learned m	JS	46.16	36.43	86.65	79.83
MetaMix, learned m	InfoNCE	45.48	35.26	86.51	79.14
DER++ + AugMix	JS	**48.99**	39.68	87.92	81.81

Table 2. Reference methods with learned or sampled original-image weight. Entries are three-seed mean accuracies (%). Bold marks the highest mean in each metric among these reference methods.

4. Discussion

The most consistent gain on corrupted class-IL is from classifier InfoNCE at stronger augmentation. Chain-aware InfoNCE attains the highest observed mean on this metric, while the advantage over MetaMix changes sign across matched strengths. The maximum is selected from five settings on the reported evaluation. With three seeds, dispersion is substantial: at $m = 0.1$ the two InfoNCE corrupted class-IL means are 43.89 ± 5.68 and 41.66 ± 7.68 (sample SD). Isolating the effect of candidate descriptors would require holding outer-view sampling and per-view weight sharing fixed. Appendix analyses report seed variation, loss differences, chain-weight concentration, and per-corruption results.

Reproducibility. The repository includes the three-seed reporting script, the one-step trainer, and the scripts for the fixed-strength sweep and AugMix coefficient comparison.

References

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A. Additional analyses

		Class-IL		Task-IL	
Mixer	m	Clean	Corrupt	Clean	Corrupt
Chain-aware	0.1	$+7.88$	$+5.69$	$+2.41$	$+2.76$
	0.3	$+7.66$	$+5.95$	$+0.31$	$+0.65$
	0.5	$+1.67$	$+0.93$	$+0.86$	$+1.78$
	0.7	$+3.14$	$+1.76$	$+0.31$	$+0.01$
	0.9	$+0.46$	$+0.77$	$+0.97$	$+1.79$
MetaMix	0.1	$+3.56$	$+2.43$	$+1.18$	$+1.29$
	0.3	$+9.36$	$+8.20$	$+0.82$	$+2.03$
	0.5	$+2.20$	$+0.89$	$+0.23$	$+0.96$
	0.7	$+3.94$	$+1.94$	$+0.95$	$+0.52$
	0.9	-1.59	-1.74	-0.02	$+0.31$

Table 3. Change in mean accuracy (percentage points) when the classifier switches from JS to InfoNCE, holding the mixer and m fixed.

Which corruptions account for the differences? Tables 6 and 7 give the results for each corruption. With InfoNCE and $m = 0.5$, chain-aware mixing exceeds MetaMix on 13 of 15 class-IL corruption means. The largest advantage is on pixelation (39.74% versus 39.10%). MetaMix leads on impulse noise and contrast. Changing the classifier loss within the chain-aware mixer produces larger differences: switching from JS to InfoNCE raises impulse-noise class-IL by 6.63 points but lowers fog class-IL by 3.57 points. When the true task is supplied, the same switch raises accuracy on 14 corruptions, with motion blur the exception (82.33% with JS and 81.72% with InfoNCE).

How concentrated are the learned weights? As the augmented share $1 - m$ increased from 10% to 90% ($m = 0.9$ to 0.1), the chain-aware mixer's final-epoch mean maximum chain weight rose at every tested step: from 33.38% to 36.04% with JS and from 33.43% to 38.24% with InfoNCE (Table 4). Thus the highest-scoring chain received a larger fraction of the augmented share as augmentation became stronger. At $m = 0.1$, chain-aware assigned more weight to its leading chain than MetaMix under both losses, while its corrupted class-IL accuracy exceeded MetaMix's only under InfoNCE (Table 1).

	Chain-aware		MetaMix	
m	JS	InfoNCE	JS	InfoNCE
0.1	0.3604	0.3824	0.3463	0.3437
0.3	0.3448	0.3672	0.3371	0.3446
0.5	0.3399	0.3426	0.3360	0.3414
0.7	0.3372	0.3358	0.3345	0.3356
0.9	0.3338	0.3343	0.3337	0.3342

Table 4. Final-epoch mean maximum chain weight, averaged equally over seeds 17, 23, and 43. We take the maximum across the three chains before averaging. Equal weights give $1/3$; the weights divide the augmented share $1 - m$.

B. Variation across training seeds

Mixer and loss	m	Clean class-IL	Corrupt class-IL	Clean task-IL	Corrupt task-IL
CA JS	0.1	43.99 ± 8.64	38.20 ± 6.12	86.05 ± 2.84	80.48 ± 1.92
	0.3	42.61 ± 5.12	36.33 ± 3.76	87.56 ± 1.01	81.29 ± 1.19
	0.5	44.59 ± 3.42	36.56 ± 3.15	86.26 ± 1.99	79.14 ± 1.77
	0.7	42.94 ± 7.54	34.96 ± 4.90	86.65 ± 1.00	79.63 ± 0.97
	0.9	42.87 ± 9.60	33.41 ± 7.11	85.58 ± 1.18	77.64 ± 1.91
CA InfoNCE	0.1	51.87 ± 7.29	43.89 ± 5.68	88.46 ± 0.72	83.24 ± 0.62
	0.3	50.27 ± 4.64	42.28 ± 3.03	87.87 ± 0.53	81.94 ± 0.29
	0.5	46.26 ± 7.46	37.49 ± 5.95	87.12 ± 1.47	80.92 ± 1.31
	0.7	46.08 ± 4.40	36.72 ± 3.01	86.96 ± 0.68	79.64 ± 1.22
	0.9	43.33 ± 6.16	34.18 ± 4.78	86.55 ± 0.63	79.43 ± 0.93
MM JS	0.1	45.69 ± 7.10	39.23 ± 4.69	86.56 ± 1.79	80.88 ± 1.08
	0.3	40.78 ± 6.02	34.31 ± 5.57	86.61 ± 2.68	79.82 ± 3.24
	0.5	43.51 ± 4.55	36.38 ± 3.30	86.89 ± 0.94	80.05 ± 0.62
	0.7	41.69 ± 4.83	34.26 ± 2.55	86.12 ± 1.35	79.07 ± 0.74
	0.9	45.60 ± 7.86	36.34 ± 4.37	86.55 ± 1.09	78.88 ± 0.86
MM InfoNCE	0.1	49.25 ± 9.09	41.66 ± 7.68	87.74 ± 1.88	82.17 ± 1.92
	0.3	50.14 ± 5.61	42.51 ± 3.66	87.43 ± 1.42	81.85 ± 1.15
	0.5	45.71 ± 8.40	37.27 ± 6.33	87.12 ± 1.80	81.01 ± 1.41
	0.7	45.63 ± 5.83	36.20 ± 3.97	87.07 ± 0.76	79.59 ± 1.31
	0.9	44.01 ± 5.66	34.60 ± 4.26	86.53 ± 0.51	79.19 ± 0.77

Table 5. Fixed-weight sweep, mean accuracy (%) $\pm$ sample SD over three seeds.

C. CIFAR-10-C accuracy by corruption

Each entry averages the five severity levels within a seed and then takes the arithmetic mean across seeds 17, 23 and 43. Overall gives equal weight to each of the 15 corruptions. Class-IL predicts among all ten classes. Task-IL predicts within the true class pair.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	28.76	**33.76**	30.49	33.61	29.49	30.42
Shot Noise	32.53	**36.57**	33.82	36.38	32.92	33.06
Impulse Noise	22.97	29.60	25.64	**29.70**	25.64	27.40
Defocus Blur	**44.85**	43.99	44.19	43.57	42.32	40.62
Glass Blur	34.00	**36.04**	34.17	36.01	35.28	34.27
Motion Blur	**41.58**	38.66	40.59	38.50	38.26	34.97
Zoom Blur	**44.66**	42.77	44.21	42.58	40.41	38.72
Snow	33.57	**36.51**	33.41	36.27	35.99	35.09
Frost	31.30	31.09	30.49	31.01	**32.48**	30.39
Fog	**39.47**	35.90	37.97	35.81	37.59	34.91
Brightness	39.68	40.77	38.47	40.34	**42.22**	40.50
Contrast	**30.16**	28.86	29.45	28.99	29.37	27.61
Elastic Transform	**44.42**	43.45	43.36	42.96	42.78	41.28
Pixelate	37.71	**39.74**	37.20	39.10	37.96	36.84
JPEG Compression	42.79	**44.68**	42.26	44.17	43.75	42.88
Overall	36.56	**37.49**	36.38	37.27	36.43	35.26

Table 6. Class-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	73.07	77.64	75.04	**77.69**	74.89	73.52
Shot Noise	76.24	80.28	77.99	**80.56**	77.69	76.28
Impulse Noise	71.88	76.32	73.34	**76.40**	73.06	72.47
Defocus Blur	84.20	84.62	84.76	**84.79**	83.76	83.58
Glass Blur	73.70	77.05	75.27	**77.34**	75.58	74.83
Motion Blur	82.33	81.72	**83.09**	82.12	81.86	80.39
Zoom Blur	83.15	83.18	**83.75**	83.39	82.30	82.18
Snow	78.97	**81.59**	79.92	81.55	79.90	80.89
Frost	78.34	**80.09**	78.96	**80.09**	79.31	79.12
Fog	80.01	80.95	80.07	**81.00**	80.57	79.14
Brightness	83.97	**84.71**	84.60	84.54	84.56	84.43
Contrast	70.05	70.82	70.28	70.71	**71.21**	68.55
Elastic Transform	83.42	84.15	84.12	**84.36**	83.61	83.65
Pixelate	83.92	**85.66**	84.60	85.55	84.50	83.69
JPEG Compression	83.90	85.01	84.98	**85.10**	84.69	84.36
Overall	79.14	80.92	80.05	**81.01**	79.83	79.14

Table 7. Task-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.